

Childhood and Community Adversity: The New Hartley Colliery Disaster 1862

When the British census was taken in April 1861, the pit village at New Hartley on England's Northumberland coast was typical of England's rapidly-expanding coal industry: a small rural enclave dominated by male breadwinning and shift work where life revolved entirely around the mine. Eight months later, the shaft of the Hester Pit coal mine in New Hartley (of Northumberland, UK) collapsed. This small community lost two hundred men and boys from carbon monoxide poisoning in a single day, leaving families and neighbours bereft. Such severe environmental stress has broad impacts on individuals and communities, including the very young. Adverse experiences affect educational, socioeconomic, and health outcomes over the life course (e.g., Gold et al. 2021; Petruccelli, Davis, and Berman, 2019). Initially, historical research into life-course adversity focused on epidemiological questions but has recently expanded into biosocial sequelae (e.g., Depauw and Oxley, 2019; Schneider, Ogasawara and Cole, 2021). In an ongoing research project supported by the society's Roger Schofield Fund, we analyse the disaster's impact on bereaved wives and children through records generated by a large-scale charitable response following the disaster, which we link to the censuses for rich longitudinal information. The Hester Pit disaster has attracted scholarly attention centered on the narrative history of the event and its immediate aftermath, as well as aspects of worker's compensation, industrial journalism, and mine safety (McKutcheon, 1963; Benson, 1974; Bronstein, 2014; Buckland, 2021; Tonks, 1999). Expanding on the demographic work of Jones (2001), our approach will investigate the specific longitudinal responses of households affected by the loss of male workers in a community heavily dependent on male breadwinning.

On a fund-supported research visit to Northumberland Archives, we focused on records created by a charitable relief fund and a school for orphaned children. These sources allow us to follow widows' and children's outcomes after the disaster at very high resolution. We also undertook preliminary work to link the bereaved families to the 1861 census. Out of the 204 men who died in the collapse, we have identified 151 in the 1861 census. From these 151 men, we count 530 surviving members of 135 bereaved households. Most lost their male head (58 per cent), and many lost sons (36 per cent). In this research note, we report preliminary evidence from our integration of census, relief, and school records.

Bereaved family members ranged in age from infancy to their sixties, and although the numerical mortality within households varied significantly, losses were drastic. The 1861 census shows us the Oliver family had seven members: 56-year-old Willam, his wife Mary (age 45), and their children John, James, William, Peter, and Ellen. All five males were killed in the 1862 disaster, with only Mary and Ellen surviving. Even in households with only one casualty, the losses were still enormous: for example, the Ainsley family—made up of John, Elizabeth and their son John—lost ‘only’ John, age nineteen.

Most were left without an adult male breadwinner in an area that offered comparatively few economic opportunities to women, who instead specialised in the extraordinarily demanding (if unpaid) shift work that mining extracted from households (Humphries & Thomas, 2023). A widow’s relief payments were not calculated according to the income her family had lost, but by her number of dependent children (of boys under fourteen and girls under sixteen); consequently, wage-replacement rates varied. Fund administrators incentivised remarriage by offering a £20 dowry. These factors allow us to pursue several further questions. How did age and gender create variation in responses to the death of wage-earning men in a community that relied on extensive household specialisation in male wage earning? How did subsequent migration choices and socioeconomic outcomes differ? Of the 88 widows we were able to preliminarily identify in the 1861 census, at least 69 remarried. The interval between disaster and remarriage varied substantially. Jane Cheetham, widow of Alfred Cheetham, remarried eight months after the disaster, while Mary Housetown, widow of Andrew Housetown, only remarried 20 years later. Next we will analyse the determinants of remarriage; this will help us identify the mechanisms through which the impact of disaster on the household was mediated.

Following the disaster, a school was opened in April 1862 to provide free education to the orphaned children, and to other local children for a fee. During the period 1862 to 1870, the school provided education for 547 children. Of these, 110 were children paternally orphaned in the 1862 disaster, with the remainder paying fees. Of the latter, at least 70 per cent were the children of fathers who worked in mining (see table 1 for a list of the fathers’ occupations). Children entered and left the school at varying ages, with the youngest student aged one and the oldest aged fourteen. We were unable to identify a school teacher in New Hartley in the 1861 census and find a large degree in the variation of ages of admission of both orphaned and non-orphaned fee-paying children, as Madoc-Jones (1977) also finds in London, which we believe

suggests that the opening of the charity school represented an improvement in local education supply. For example, the surviving children of the bereaved Amour family, who lost father and eldest son in the disaster, all entered the school in its first week aged five, eight, ten, and 12. The median admission age of children in both groups was identical at six years, and the mean ages of admission were not statistically significant different at 6.9 and 6.5 years respectively.

The school's daily attendance registers note childrens' frequency and cause of absence across nine years. Our preliminary analysis suggests that in the first year of its opening, fee-paying children attended less regularly than paternal orphans, with the former group attending 44 percent of potential school sessions and the latter attending 60 percent of potential sessions. We believe this affirms the importance of income constraints in restricting school attendance before the elimination of elementary school fees in the later 19th century. We plan to analyse attendance patterns during developmentally-sensitive periods of childhood (Tomer, 2015), as well as the influence of gender, household demography and income loss, and parental resources upon children's school attendance and later outcomes; here we will build on a literature that has identified human capital as an important channel through which adverse childhood experiences impact later outcomes (Schurer et al., 2019). Planned analysis will decompose these differences in attendance by reported cause of absence, whether for illness, caring responsibilities, inability to pay, or other reasons, and further estimate whether these differences in attendance translate into differences in later-life outcomes for bereaved children. Studying these dynamics will allow us to test for sensitive periods in human capital formation. Methodologically, combining historical school attendance records with longitudinally-linked census microdata promises to be a fruitful direction, providing insights into the complex links between household responses to shocks and individual outcomes over the lifecourse. This will improve our understanding of children's roles in familial (re)productive labour, the prevalence and seasonality of morbidity over the lifecourse, and gender inequalities.

The 1862 Hester Pit disaster fractured a community, and by uncovering how individuals and households coped our study sheds light onto gendered labour in coal mining and human capital formation. Moreover, by linking the numerous sources created following the disaster with the censuses, promises to advance methodologies for historical life course reconstructions.

Table 1: Occupations of fathers of children admitted to school:

	Freq.	Percent
BLACKSMITH	5	1.34
BRAKESMAN	3	0.81
BUTCHER	6	1.61
CARTMAN	7	1.88
COKE BURNER	3	0.81
COLLIER	111	29.84
COUNTRYMEN	1	0.27
FARMER	6	1.61
FATHERLESS	15	4.03
HEWER	3	0.81
HIND	7	1.88
JOINER	3	0.81
LABOURER	1	0.27
MASON	5	1.34
MINER	75	20.16
ORPHAN	110	29.57
PLATELAYER	2	0.54
POLICEMAN	1	0.27
SINKER	2	0.54
STATION MASTER	2	0.54
WEIGHMAN	3	0.81
WIDOW	1	0.27

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